# Interface Design after Monolith Decomposition during the Migration to Microservices: A Systematic Literature Review

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***Abstract*—Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum.**

## I. Introduction

**NOTES:**
Traditionally, teams in Software Engineering of business applications are divided along technical capabilities.

According to Conway's Law, which is quoted in Quote 1, the technically oriented team decomposition yields to the technically oriented decomposition of the software output, which is illustrated in figure Fig. 1.

> "Any organization that designs a system (defined broadly) will produce a design whose structure is a copy of the organization's communication structure."

Quote 1. Conway's Law quoted by Melvin Conway, 1968 [1]

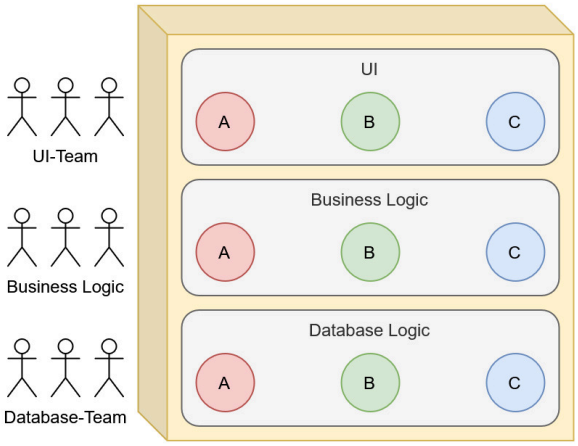



Fig. 1. Illustration of a business application built as a monolith. Original illustration from C. Nockemann [2].

## II. Background and Related Work

### A. Monolith vs. Microservices

### B. Migration

a) *Migration Phases:*

**NOTES:**
b) *Migration Strategies (according M. Abdellatif* et al. *[3]):* The authors M. Abdellatif *et al.* [3] suggest three general strategies of migrating a legacy system into a SOA: *top-down-*, *bottom-up-* and *hybrid strategies*

c) *Domain-Driven Design (DDD):* < **TODO: DDD Buch verwenden und referenzieren** >

- tactical design and strategiel design
- tactical design defines how the domain model is composed < **TODO: Beispiel einbauen?** > and to identify bounded Contexts
- strategiel design defines how the bounded contexts communicates

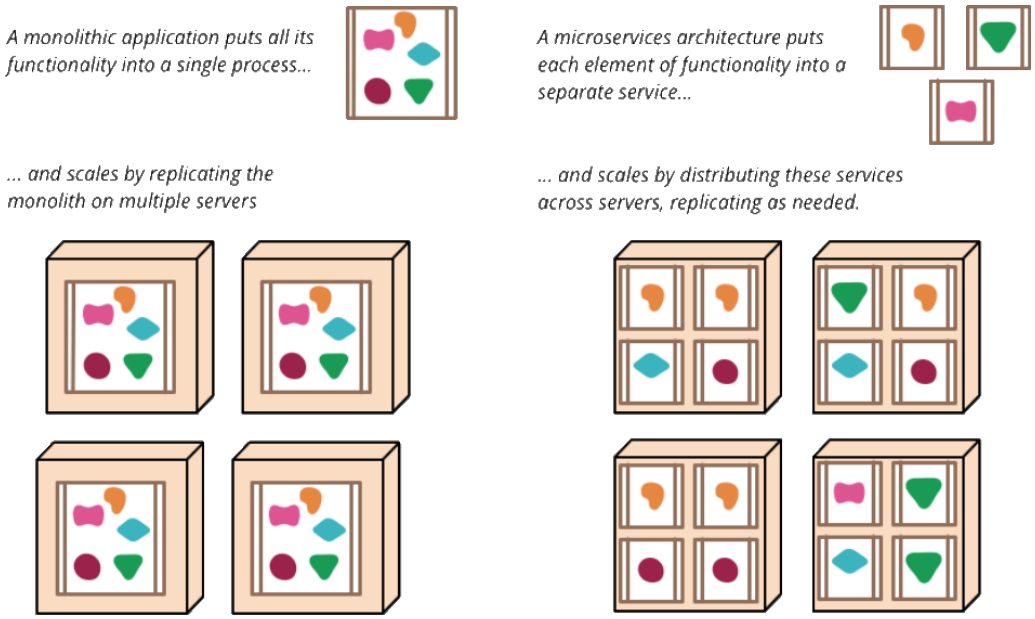



Fig. 2. Scaling and deploying of Monoliths and Microservices by J. Lewis and M. Fowler [1]

### C. *Microservice Interfaces*

### D. *Existing SLR's*

## III. Methods

**< TODO: Mehr in SLR-Methodik einlesen >**

**NOTES:**

### A. *Planning Phase*

- Identify the need for a review and the development of a review protocol
  - leading to the development of research questions

a) *Identify the Need for an SLR:*

- Explain why your SLR is necessary?
- Which gap does it close?

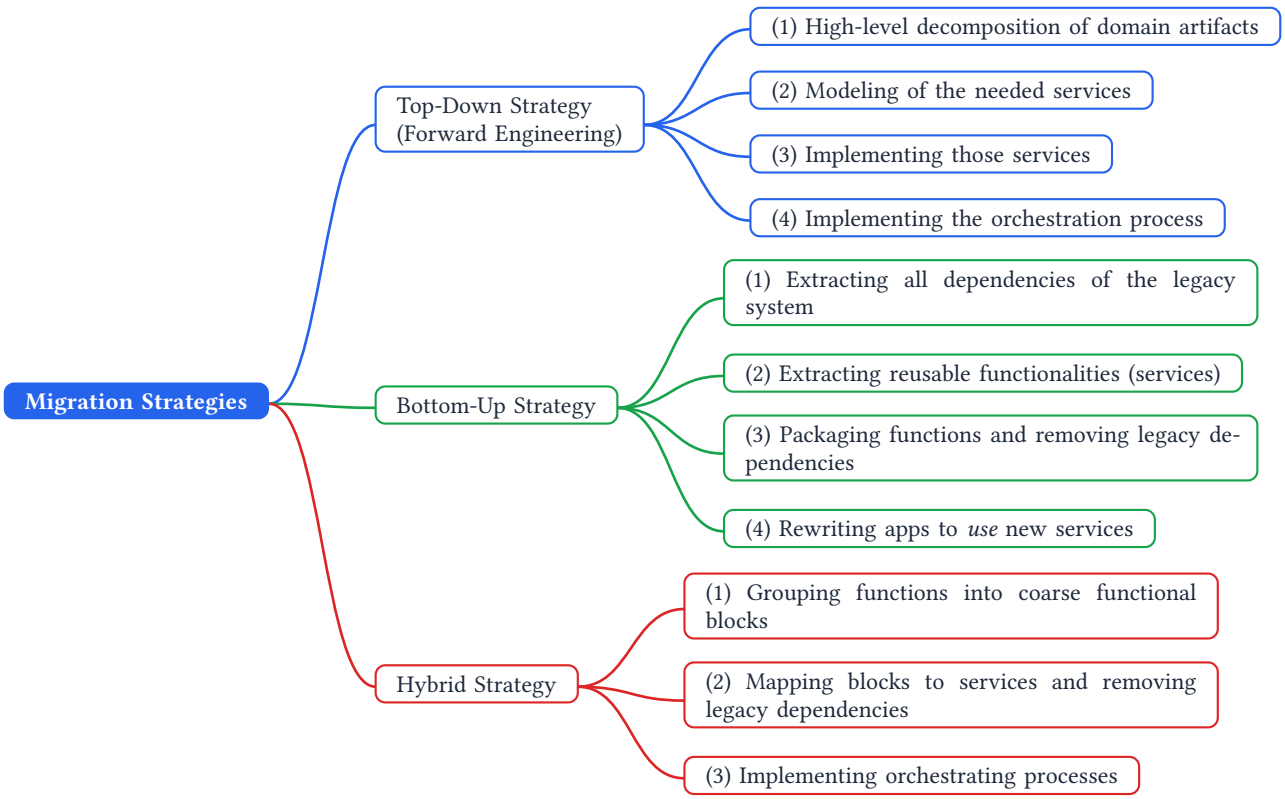



Fig. 3. Own created mindmap of the authors M. Abdellatif *et al.* [3] proposed migration strategies.

- Which other Literature does exist (only coarse, in a nutshell, not detailed)?
- Why would be an SLR beneficial for your area?

b) *Specifying the Research questions:*

- What are your research questions?

c) *Defining and Evaluating the Review Protocol:*

- Where and who has performed the SLR? Which university?
  - Who lead the SLR, who wrote the protocol and who performed the Evaluations?
- What are your IC and ECs?

### B. Review Phase

a) *Selection of Primary Studies:*

- Talk about your identified Keywords.
  - What were your initial keywords? What synonyms did you found?
- Which Search Strings did you used?
- Explain how you selected the studies. Which inner phases and refinements the studies went through?
- What happened during the semester? Which meetings, how often you have meet with the supervisor?
- Explain *Snowballing*. What iterations did you do applying snowballing?
- How many Paper are included thank to snowballing
- Provide a figure showing the progress the paper went through and show how many are discarded in each step.
- Provide a table with all selected studies, including title, number, Type, Year and so on...

b) *Analysis of the Data:*

- Explain how you extracted the Data. Which Tools did you use?
- What Data did you extract?
- Which data was obvious and inviolable?

## IV. Results

**NOTES:**
Present the Results of your Literature Review

### A. Stats over the selected studies

a) *Temporal distribution of the studies:*

- Provide a stats about paper per year

b) *Publication venue:*

- Which are the (Main-) Conferences, Journals usw...?
- Which are the search platforms?

## V. Discussion

## VI. Conclusion

## VII. Appendix

### A. Searched Databases with Search Strings

- IEEE Xplore Digital Library
  - "microservice*" AND "identif*" AND ("interface" OR "api")

### B. Searched Journals

- Information and Software Technology
- Journal of Systems and Software
- IEEE Transactions on Software Engineering
- Computer Applications in Engineering Education

### C. Searched Conferences

- International Conference on Smart Applications and Data Analysis (SADASC)
- International Workshop on Software Engineering Aspects of Continuous Development and New Paradigms of Software Production and Deployment (DEVOPS)
- International Conference on Software Engineering and Formal Methods (SEFM)
- International Conference of the Chilean Computer Science Society (SCCC)
- Second International Conference on Information Systems and Software Technologies (ICI2ST)
- IEEE International Conference on Web Services (ICWS)
- European Conference on Service-Oriented and Cloud Computing (ESOCC)
- International Conference on Web Engineering (ICWE)
- International Conference on Evaluation and Assessment in Software Engineering (EASE)
- IEEE International Conference on Software Architecture Workshops (ICSAW) renamed since 2019 to IEEE International Conference on Software Architecture Companion (ICSA-C)
- Open Conference of Electrical, Electronic and Information Sciences (eStream)
- International Conference on Information Technology-New Generations (ITNG)

## References


[1] J. Lewis and M. Fowler, "Microservices: A Definition of This New Architectural Term." Accessed: Aug. 25, 2026. [Online]. Available: https://martinfowler.com/articles/microservices.html

[2] C. Nockemann, "Domain-Driven Design in Der Praxis," in *Presentations during the Lecture of Object-Oriented Component Architectures (OOKA) Renamed to Service-Oriented Component Architectures (SEKA)*, University of applied science Bonn-Rhein-Sieg, 2025.

[3] M. Abdellatif *et al.*, "A Taxonomy of Service Identification Approaches for Legacy Software Systems Modernization," *Journal of Systems and Software*, vol. 173, p. 110868, Mar. 2021, doi: 10.1016/j.jss.2020.110868.